A Gain Is Not Proof of Learning: SkillRC, a Reality Check for Distilled Agent Memory

Yiwen Lu
University of Notre Dame
ylu37@nd.edu

Research report, November 2026. Code and data: https://github.com/EvenEureka/SkillRC

Abstract

A gain over no memory is the usual evidence that an agent has learned from experience, yet for memories distilled from past runs into rules placed in the prompt it does not show that learning happened, because a memory also adds text to read and is one draw from the pipeline that built it. We propose SkillRC, a reality check that compares a memory with matched controls, which keep its items, order and token count and replace its content, and repeats the comparison over pools built by the same pipeline. With six language models on ALFWorld and WebShop, a memory's lead over unrelated text is modest, 1.2 to 5.0 points for four local models over many pools. Over three pools, distilled rules lead generic rules from the same writer by 3.5 points but advice for another environment by only 1.1 [−2.7, 4.9]. Replicate pools differ about as much as the value itself, with a standard deviation of 1.9 points that includes unseparated run variation, and one WebShop pool lowers scores by 11.6 to 15.8 points against its control. Adding the control also changes what is detected: of 16 net gains on single pools, 8 come with no detected content gain, although this does not show that they are false. A gain on one memory is therefore weak evidence of learning, and the unit worth evaluating is the pipeline that builds memories.

1 Introduction

The usual evidence that an agent has learned from experience is a gain over the same agent without memory, and such gains are uneven. A common way to build such a memory is to distill successful runs into a short list of rules placed in the prompt (Zhao et al., 2024; Shinn et al., 2023; Wang et al., 2025). Evaluations of agent skills find gains that vary across tasks and models and are sometimes negative (Li et al., 2026; Han et al., 2026), and that shrink when skills have to be retrieved under realistic conditions (Liu et al., 2026). In our own runs, a list of 33 rules that we chose because it had helped GPT-4o-mini raised that model's success on ALFWorld by about 9 points and changed the success of four other models by less than 3. We ask what such a gain shows: did the agent learn from experience?

A gain over no memory cannot answer this, for two reasons. First, adding a memory changes two things at once: the model has more text to read, and the text says something about the task. Language models are sensitive to irrelevant context and to the length and format of their input (Shi et al., 2023; Levy et al., 2024; Sclar et al., 2024), so the first change alone could move success by as much as the gains being reported. Second, one list is one draw from the pipeline that built it, and a verdict on that list may not hold for the next list the pipeline builds. Evaluations so far vary the tasks and the models, and at most partly control for the act of reading.

To tell these apart, we compare a memory with matched texts and repeat the comparison over pools, a design we call SkillRC (Figure 1). A matched control keeps the memory's shape and replaces its content: the same items with the same tags and word counts, in the same order, with the same token count under each tested model's tokenizer, but with each rule replaced by a sentence about vegetable gardening. The control minus no memory gives the *reading effect*, the change when the prompt gains a block of this shape with unrelated text. The memory minus the control gives the *content value*, the gain of the rules over that text. These contrasts are relative to the chosen control, and they do not identify a topic-independent reading cost or an experience-specific learning effect. SkillRC then asks what this value is made of, by comparing the memory with rules written without experience and with advice for another environment, and how far it holds, by repeating the comparison over several pools built by the same

pipeline. SkillRC is an evaluation design and does not aim to raise success, and our setting is short rule lists placed whole in the prompt.

We apply SkillRC on ALFWorld (Shridhar et al., 2021) and WebShop (Yao et al., 2022) with six models, and what remains of a gain is modest and unstable. Over many pools, the memory leads unrelated text by 1.2 to 5.0 points for four local models. Over three pools it leads generic rules from the same writer by 3.5 points but advice for another environment by only 1.1. Pools built the same way differ about as much as the value itself, and one WebShop pool lowers scores by 11.6 to 15.8 points against its control. These results do not show that distilled memory is worthless, since for two readers rules beat their own scrambled words. They show that a gain on one memory is weak evidence of learning.

Matched comparisons reveal control-sensitive gains and variation across replicate pools that a single net-gain estimate conceals. Matched controls and averaging are established tools, and we apply them to experience memory with a match verified on every task, comparison texts and replicate pools (Section 3). With this design we find modest contrasts for four local readers, sensitivity to the comparison text for two Qwen3 readers, replicate-pool variation for three Qwen3 readers and negative contrasts for two WebShop readers (Sections 5.1–5.5). Adding the matched control also changes nominal gain detection in 11 of 54 evaluations, without establishing which detections are false discoveries (Section 5.4). For claims that a gain comes from experience, we recommend evaluating the pipeline that builds memories over several pools and more than one control.

2 Related Work

Experience memory and its evaluation. Agents store experience as executable skills (Wang et al., 2024), verbal reflections (Shinn et al., 2023), distilled insights (Zhao et al., 2024; Fu et al., 2024), reusable workflows (Wang et al., 2025) and evolving playbooks (Zhang et al., 2026), and later systems curate skills around a fixed executor (Yang et al., 2026; Ouyang et al., 2026; Gao et al., 2026) or optimize the whole harness in which a memory sits (Lee et al., 2026; Huang et al., 2026). Most of this work supports its methods with a gain over the same agent without memory. Evaluations that look across tasks and models find such gains unstable. SkillsBench reports negative deltas on some tasks and notes that an injected skill also adds context, using self-generated skills as a partial control (Li et al., 2026), and SWE-Skills-Bench finds limited gains with a token overhead (Han et al., 2026). SkillEvoSkill compares executions with and without a skill to revise the skill (Gao et al., 2026); we use the same kind of matched comparison to evaluate a fixed memory, adding a control for the act of reading. These studies show that the effect of a memory varies, without separating the part due to reading added text.

Controls for added context. Prompting research has made that kind of separation for a long time. Content-free inputs calibrate few-shot predictions (Zhao et al., 2021), irrelevant instructions can work about as well as good ones (Webson and Pavlick, 2022), and demonstrations with random labels keep much of their benefit (Min et al., 2022). On ALFWorld itself, ReAct performance is driven by the similarity between demonstrations and queries more than by the content of the reasoning traces (Verma et al., 2024). These controls were designed for a single prompt. An agent's memory is reread at every step of an episode by models whose tokenizers differ, and it comes from a pipeline that can produce a different list each time. SkillRC therefore differs in three respects: its control is built for one specific memory, the match to what the harness inserts is audited on every task for each reader, and the comparison is repeated over pools from one pipeline. We have not found a study that combines control-relative contrasts for agent memory across readers with replication over pools from one pipeline.

3 Method

SkillRC runs the same tasks under three conditions that differ only in one block of the prompt: a memory, a matched control or nothing (Figure 1). A memory is a pool of short rules, each tagged with a task type and distilled from successful trajectories. We call the frozen model together with its tokenizer, chat template and serving the *reader*. Section 3.1 defines the two quantities, Section 3.2 builds the control that separates them, and Section 3.3 explains why one comparison is not enough.

3.1 Reading effect and content value

A comparison with and without a memory changes both the amount of text and what it says, so telling

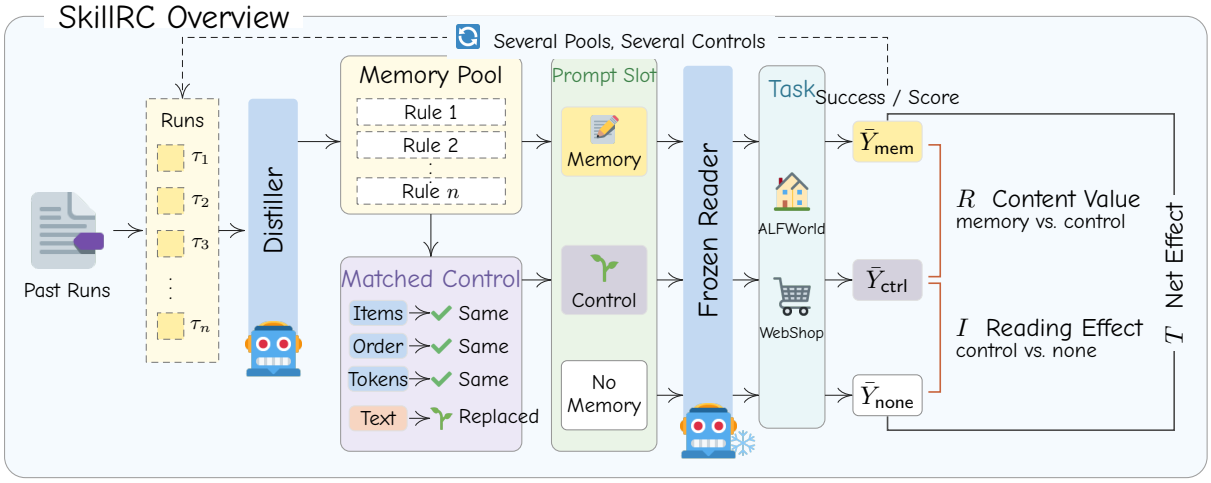


Figure 1: Overview of SkillRC. One frozen reader runs the same tasks three times, and only the text in one prompt slot changes: a distilled memory, a matched control with the memory's items, order and token count but replaced text, or nothing. The gaps between the three success rates give the content value R and the reading effect I, which add up to the net effect T that a with-and-without comparison reports. The audit is repeated over several pools and several controls.

them apart takes a third condition that keeps the text's shape and replaces what it says. Let $\bar{Y}_{\text{none}}$, $\bar{Y}_{\text{mem}}$ and $\bar{Y}_{\text{ctrl}}$ be a reader's mean success without memory, with the memory and with the control. The net effect splits as

$$\underbrace{\bar{Y}_{\text{mem}} - \bar{Y}_{\text{none}}}_{T,\text{ net effect}} = \underbrace{\bar{Y}_{\text{ctrl}} - \bar{Y}_{\text{none}}}_{I,\text{ reading effect}} + \underbrace{\bar{Y}_{\text{mem}} - \bar{Y}_{\text{ctrl}}}_{R,\text{ content value}}. \quad (1)$$

Equation 1 is an algebraic identity, and its causal interpretation requires an unverified assumption about the topic of the control. T is what a with-and-without comparison reports. A negative I means that a block of unrelated text lowers success, and a positive R means that the rules do better than that text in the same place. Reading I as the effect of reading and R as the effect of content assumes that the effect of reading does not depend on the topic of the text, which our experiments do not establish. Section 5.2 shows how far the estimates move with a different control. We keep the names reading effect and content value as labels for these two contrasts.

3.2 Matched control

Length, format and position all change what a model does, so the control has to match the memory in each of them. For each tested reader, we call two conditions *payload-matched* when they insert corresponding items in the same order and with equal token counts on every task. Given a memory, we first keep each item's tag and word count and replace its body with an imperative sentence about vegetable gardening, drawn from a bank screened against a vocabulary exclusion list; the first control retains some incidental overlap, including the word "fork". For example, "Always clean the item using the designated cleaning area before placing it in the final location" becomes "Always pull weeds while they are small so they never set seed across the whole garden", which also has 16 words. We then rank the control with the memory's BM25 scores (Robertson and Zaragoza, 2009), so that each item lands in the same position. Finally, a greedy pass, deterministic first and then with seeded orders, replaces body words with the nonce token x until the block has the memory's token count, without guaranteeing the minimum number of replacements.

Because the match is defined on what the harness actually inserts, it can be verified. The harness logs the token count in every episode, and an offline audit replays retrieval on every task to confirm the items and their order.

3.3 Comparison texts and replicate pools

The matched control shows what the rules add over unrelated text, which leaves open whether that addition comes from experience. Two further texts of the same shape address this. Generic rules are written by the same model without seeing a trajectory, so they show what the writer can say without experience. Advice for another environment keeps a plausible rule-list format but may contain transferable strategies, so this comparison tests a

task-specific advantage over that advice.

A pool is also one draw from a pipeline, so a value measured once could reflect the draw more than the method. SkillRC therefore reports estimates pooled over replicate pools, which keep the source agent, the distiller and the number of trajectories per task type and change only the seed that selects the trajectories. This costs more runs: with k pools and one control an audit needs $2k + 1$ runs per reader, against 2 for one with-and-without comparison, so eight pools take 17 runs, and each further control adds k.

All estimates are differences of paired task means. Intervals are 95% percentile intervals from 20,000 bootstrap draws of tasks within task type (product category on WebShop), also resampling subsamples within source for the crossover and individual pools or reader-pool cells for the same-window pooled analyses.

4 Experimental Setup

Environments. On ALFWorld we use the 134 unseen games with a ReAct scaffold (Yao et al., 2023) that lists the admissible actions, two demonstrations per prompt and a limit of 30 steps; each run covers every game with three demonstration pairs, 402 episodes. WebShop replaces rooms and objects with search and product pages, which tests whether the decomposition carries over to another action space. We use the 1,000-product environment with a seeded server and 500 test sessions of at most 15 steps, scored in $[0, 1]$.

Readers. The six readers span a range of ability, from 26% to 90% success on ALFWorld without memory. GPT-4o-mini and GPT-4.1-mini are called through the OpenAI API. Llama-3.1-8B (Grattafiori et al., 2024) and Qwen3-8B, Qwen3-14B and Qwen3-32B (Yang et al., 2025), with thinking off, are served locally with vLLM (Kwon et al., 2023). All decode at temperature zero.

Memories. Our main evaluation is a crossover at balanced coverage. Each of three source agents (GPT-4o-mini, Qwen3-8B, Qwen3-32B) supplies the same number of successful trajectories per task type, three replicate subsamples per source, and each subsample is distilled once by Qwen3-32B and once by Qwen3-8B. ALFWorld trajectories come from the 140 seen games and WebShop's from 576 training sessions, and evaluation uses separate test sets. Subsamples from one source can overlap, because a source has a limited number of successes. This gives 18 pools of 31 to 36 rules, each with its own gardening control, and the three Qwen3 readers read all of them, for 54 evaluations. Llama-3.1-8B reads six of these pools. GPT-4o-mini reads the 33-rule pool from the introduction, distilled from its own successes. On WebShop one pool of 30 rules, distilled from GPT-4o-mini's successes, is compared with a control about choral singing, because WebShop sells garden products.

Baselines. The baseline evaluation is the net effect T on one pool. On three pools in one time window, Qwen3-8B and Qwen3-14B also read two further texts of the same shape: generic rules that Qwen3-32B, the writer of these pools, wrote without seeing a trajectory, and advice for another environment, rephrased from our WebShop rule pools screened against the control builder's exclusion vocabulary, which exempts stopwords.

Protocols. These evaluations come from six studies whose predictions, estimands and analyses were written down before the runs, except for one addendum written after two hosted readers had been analysed. Together the studies cover 24 ALFWorld pools. We mark everything chosen after seeing data as exploratory. Repeating a run under identical settings changes 17–20% of GPT-4o-mini episodes and 4–14% of local-reader episodes, so differences of about 2 points between single runs should not be over-read.

5 Results

5.1 Against unrelated text, a memory's value is modest

A memory's lead over unrelated text is small for most readers (Table 1). Over many pools the content value is $+5.0$ points for Llama-3.1-8B and $+3.9$ for Qwen3-8B, and for Qwen3-14B and Qwen3-32B, at $+1.3$ and $+1.2$, it cannot be distinguished from zero. The largest value, $+9.1$, is GPT-4o-mini's on a single pool built from its own successes, the pool we chose because it had helped that model.

A net effect can hide this in either direction. Over six pools, gardening text scores 4.4 points below no memory for Llama-3.1-8B and the memory scores 5.0 above gardening, for a net of $+0.6$, with no detectable spread between pools, so a with-and-without comparison would report nothing for this reader. An exploratory comparison on three

Table 1: What a with-and-without comparison reports (top) and what the decomposition adds (shaded), in points of success ($\times 100$; WebShop: score), with 95% intervals. Bold marks an interval that excludes zero. Llama-3.1-8B: six pools. Qwen3 readers: 18 pools. GPT-4o-mini on ALFWorld: one pool, mean of two runs. WebShop: one pool.

	ALFWorld					WebShop	
	Llama-3.1-8B	Qwen3-8B	Qwen3-14B	Qwen3-32B	GPT-4o-mini	GPT-4o-mini	Qwen3-8B
Net effect T (with vs. without)	+0.6 [−2.9, 4.0]	**+4.5** [0.9, 8.0]	**+3.3** [0.03, 6.8]	+0.2 [−2.1, 2.7]	**+8.8** [5.0, 12.8]	**−14.0** [−17.4, −10.6]	**−13.2** [−17.1, −9.3]
Reading effect I (ours)	**−4.4** [−8.2, −0.7]	+0.6 [−2.9, 4.2]	+2.0 [−1.4, 5.5]	−1.0 [−3.8, 1.8]	−0.2 [−3.7, 3.1]	+1.8 [−1.2, 4.8]	−1.6 [−5.6, 2.4]
Content value R (ours)	**+5.0** [2.2, 8.0]	**+3.9** [1.2, 6.5]	+1.3 [−0.8, 3.4]	+1.2 [−0.8, 3.3]	**+9.1** [5.0, 13.3]	**−15.8** [−19.1, −12.5]	**−11.6** [−15.3, −7.9]

Table 2: Ablation: the value that each version of the evaluation reports for the pools it covers, in points of success. Many pools: six for Llama-3.1-8B, 18 for Qwen3, against gardening text. Generic rules come from the same writer as the pools, and other-domain advice is written for WebShop. The highlighted row is the matched gardening control over many pools.

Version	Llama-3.1-8B	Qwen3-8B	Qwen3-14B
Gardening control	+5.0	+3.9	+1.3
No control	+0.6	+4.5	+3.3
One pool (range)	[+3.7, +6.5]	[−1.5, +10.4]	[−2.0, +6.2]
Three pools, one time window: memory against			
Gardening text	–	+5.4	+0.8
Generic rules	–	+5.1	+1.9
Other-domain advice	–	+1.7	+0.4

pools in one window finds the same pattern for Qwen3-8B, with shifts of -2.8 and $+5.4$, while for Qwen3-14B neither shift is distinguishable from zero. Similar net effects can also have different sources: Qwen3-14B's net of $+3.3$ looks close to Qwen3-8B's $+4.5$, yet only Qwen3-8B's has a content value distinguishable from zero.

Llama-3.1-8B's two shifts reuse the same six pool-specific gardening runs, so they are statistically dependent: gardening scored below both other conditions. Either reading any added block costs something and the content repays it, or off-topic text costs what on-topic text does not. The first reading allows a benefit offset by a reading cost, and the second allows the contrast to arise from harm caused by the unrelated control. The comparison texts in the next section bear on this for the Qwen3 readers.

5.2 Ablation: the value depends on the comparison

The reported estimate depends on the comparison condition and on which pools are averaged (Table 2). Without a control, Llama-3.1-8B's $+5.0$ reads as $+0.6$ and Qwen3-14B's $+1.3$ reads as $+3.3$. With one pool in place of many, a single estimate for Qwen3-8B falls anywhere from -1.5 to $+10.4$. These rows show how sensitive the reported value is, and they do not show which version is correct.

Changing the comparison text shows how little of the value is clearly about experience. Over the same three pools and both readers, the memory leads gardening by $+3.1$ $[-0.8, 7.7]$, generic rules from the same writer by $+3.5$ $[0.2, 7.2]$, as we predicted, and advice for another environment by only $+1.1$ $[-2.7, 4.9]$, against our prediction that it would beat that advice as well. We detected no advantage of the generic rules over gardening (-0.4 $[-3.0, 2.3]$), so the lead over them may reflect a weak generic list as much as a strong memory, and one reader-pool cell supplies nearly half of it. Sentences such as "Scan listing title on the page for the requested size before clicking at all" may carry a general habit of checking before acting, or may simply prompt care, and our data cannot yet separate those. For the Qwen3 readers this leans toward the second reading of the previous section, since the memory's lead over text on another task is small and uncertain.

The rules still matter, while the origin of the experience shows no detected effect. Replacing the rules with their own words in scrambled order lowers success by 5.3 points for GPT-4o-mini and 8.0 for Qwen3-8B, so these readers respond to more than the rules' vocabulary. We detected no effect of which agent supplied the experience (-0.2 $[-2.4, 1.9]$, Qwen3-8B against GPT-4o-mini) or of which model distilled it ($+0.5$ $[-1.2, 2.2]$).

5.3 Pools built the same way differ as much as the value

A memory's value does not carry over reliably to the next pool from the same pipeline. For a fixed

Table 3: Baseline against SkillRC. Upper rows: gains detected on single pools by the net effect (baseline) and by the content value against the matched control, for three Qwen3 readers on ALFWorld; detection means a 95% task interval above zero. Lower rows: mean absolute gap between two content-value estimates of one reader, with one pool each (baseline) and eight pools each.

	Baseline	SKILLRC (ours)
Adding the matched control: 54 one-pool evaluations		
Gains detected	16	11
Gains detected by both designs	8	8
Gains only this design detects	8	3
Adding pools: gap between two estimates, in points ↓		
Qwen3-8B (one pool vs. eight)	3.8	**1.8**
Qwen3-14B (one pool vs. eight)	2.4	**0.8**
Qwen3-32B (one pool vs. eight)	2.3	**0.7**

source, distiller and reader, replicate pools differ by more than task sampling explains ($Q = 72.2$ on 36 degrees of freedom, $p = 0.0003$), with a between-pool standard deviation of 1.9 points, about the size of the content values themselves. For Qwen3-8B, three pools distilled by Qwen3-32B from GPT-4o-mini's successes give 0.0, $+6.0$ and $+10.4$. For the net effect and the content value alike, a verdict matches the verdict on a replicate pool in only 34 of 54 pairs. The 1.9-point estimate removes task-sampling variance but does not separately estimate run-to-run variation, and replicate pools can share trajectories.

5.4 What adding a control changes

Half of the net gains that a with-and-without comparison detects come with no detected content gain (Table 3). Of the 16 evaluations with a net gain at the nominal 95% level, 8 detect no content gain against the matched control, so the comparison alone would credit the rules where the control does not. Another 3, all for Qwen3-32B, detect a content gain that the net effect does not show. Adding the control therefore changes the detection verdict in 11 of 54 evaluations. These are changes in detection and not demonstrated false discoveries: a nondetection does not establish that a value is zero, and after Holm correction over the 54 evaluations only 1 net gain and 3 content gains remain, so a verdict from one pool is weak evidence in either direction.

Averaging over pools reduces the disagreement that one pool leaves. Each eight-pool side averages both distillations of four subsamples, chosen from the nine available across three sources. Two estimates of one reader differ by 2.3 to 3.8 points with one pool each and by 0.7 to 1.8 with eight, a cut of about 54 to 69%, roughly what averaging predicts. Eight-pool sides use different subsamples, but shared trajectories average 38% of the smaller side, so these gaps are partly optimistic. Comparisons across sources share no trajectories and point the same way: over the three pairs of sources, the mean gap per reader is 0.2 to 2.2 points for six pools each, against 2.3 to 3.7 for single pools. This comparison is descriptive, because its two sides come from different sources.

5.5 A distilled memory can lower scores

On WebShop the one pool we tested lowers scores relative to both no memory and its matched control (Table 1). The pool takes GPT-4o-mini's score from 29.4 to 15.4 while the choral control leaves it at 31.2, a content value of -15.8 $[-19.1, -12.5]$, and for Qwen3-8B the content value is -11.6 $[-15.3, -7.9]$. We had predicted a positive value for GPT-4o-mini. An exploratory count of actions suggests a mechanism: the pool raises GPT-4o-mini's searches per session from 2.44 to 5.69, against 2.35 with the control, and several rules tell the agent to search again. This result comes from one pool in a harness whose observations do not name the current page, so it holds for that observation format.

6 Conclusion

A gain over no memory is the usual evidence that an agent learned from experience, and it is weak evidence. SkillRC compares a memory with a control that keeps its shape and replaces its content, adds comparison texts, and repeats over pools built by the same pipeline. Against unrelated text the value was modest, 1.2 to 5.0 points for four local readers over many pools, with intervals that include zero for two of them. Over three pools its lead over advice for another environment was 1.1 points, pools from one pipeline differed about as much as the value, and one WebShop pool lowered scores. Adding the control also changed detection: 8 of the 16 detected net gains came with no detected content gain. These results do not show that distilled memory is worthless: for two readers rules beat their own scrambled words, and on three pools they beat generic rules from the same writer. They show that a gain on one memory does not

establish learning. For claims that a gain comes from experience, we recommend reporting a memory against at least two matched controls and over several pools, and treating the pipeline that builds memories as the unit of evaluation. The extra runs buy interpretability, and any benefit for decision accuracy remains to be shown.

Our evidence has clear limits. We have no ground truth for a memory's value, so we can show that verdicts change and that estimates agree better, and not that decisions made with SkillRC are more accurate. The memories are rule lists of 520 to 790 tokens placed whole in the prompt, so retrieval never had to choose among items, and our readers are small open models and two small hosted models. The decomposition rests on the assumption that the effect of reading does not depend on the topic of the text, and replacing gardening with a second unrelated topic, amateur astronomy, moved the control's score by -2.5 points $[-4.4, -0.7]$. The comparison texts ran on three pools and two readers, so the small lead over advice is a nondetection and not a finding of no value. The two hosted readers read only a control with placeholder tokens, and the generic rules all derive from one base list, so the lead over them is a lead over this list. Intervals exclude variation across runs and time windows, the crossover shares one no-memory run per reader, and the six Llama-3.1-8B pools derive from three subsamples. The next step is to test long procedural skills under a binding retrieval budget, where reading effects may be larger, and advice for a third domain, which would show how far the value of general advice extends.

References

Yao Fu, Dong-Ki Kim, Jaekyeom Kim, Sungryull Sohn, Lajanugen Logeswaran, Kyunghoon Bae, and Honglak Lee. 2024. AutoGuide: Automated generation and selection of context-aware guidelines for large language model agents. In *Advances in Neural Information Processing Systems*, volume 37, pages 119919–119948.

Haowen Gao, Haoran Chen, Can Wang, Shasha Guo, Liang Pang, Zhaoyang Liu, Huawei Shen, and Xueqi Cheng. 2026. SelfEvoSkill: Paired execution audits for skill revision without external outcome supervision. *Preprint*, arXiv:2606.14239.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, et al. 2024. The Llama 3 herd of models. *Preprint*, arXiv:2407.21783.

Tingxu Han, Yi Zhang, Wei Song, Chunrong Fang, Zhenyu Chen, Youcheng Sun, and Lijie Hu. 2026. SWE-Skills-Bench: Do agent skills actually help in real-world software engineering? *Preprint*, arXiv:2603.15401.

Yue Huang, Wenjie Wang, Han Bao, Yuchen Ma, Xiaonan Luo, Yi Nian, Haomin Zhuang, Zheyuan Liu, Yue Zhao, and Xiangliang Zhang. 2026. MemoHarness: Agent harnesses that learn from experience. *Preprint*, arXiv:2607.14159.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, pages 611–626.

Yoonho Lee, Roshen Nair, Qizheng Zhang, Kangwook Lee, Omar Khattab, and Chelsea Finn. 2026. Meta-Harness: End-to-end optimization of model harnesses. In *Conference on Language Modeling (COLM)*.

Mosh Levy, Alon Jacoby, and Yoav Goldberg. 2024. Same task, more tokens: the impact of input length on the reasoning performance of large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15339–15353, Bangkok, Thailand. Association for Computational Linguistics.

Xiangyi Li, Yimin Liu, Wenbo Chen, Bingran You, Zonglin Di, Yifeng He, Shenghan Zheng, Kyoung Whan Choe, Jiankai Sun, Shuyi Wang, Chujun Tao, Binxu Li, Xuandong Zhao, Hejia Geng, Xiaojun Wu, Junwei Zhou, Xiaokun Chen, Hanwen Xing, Yubo Li, et al. 2026. SkillsBench: Benchmarking how well agent skills work across diverse tasks. *Preprint*, arXiv:2602.12670.

Yujian Liu, Jiabao Ji, Li An, Tommi Jaakkola, Yang Zhang, and Shiyu Chang. 2026. How well do agentic skills work in the wild: Benchmarking LLM skill usage in realistic settings. In *Conference on Language Modeling (COLM)*.

Sewon Min, Xinxi Lyu, Ari Holtzman, Mikel Artetxe, Mike Lewis, Hannaneh Hajishirzi, and Luke Zettlemoyer. 2022. Rethinking the role of demonstrations: What makes in-context learning work? In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 11048–11064, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Siru Ouyang, Jun Yan, Yanfei Chen, Rujun Han, Zifeng Wang, Bhavana Dalvi Mishra, Rui Meng, Chun-Liang Li, Yizhu Jiao, Kaiwen Zha, Maohao Shen, Vishy Tirumalashetty, George Lee, Jiawei Han, Tomas Pfister, and Chen-Yu Lee. 2026. SkillOS: Learning skill curation for self-evolving agents. *Preprint*, arXiv:2605.06614.

Stephen Robertson and Hugo Zaragoza. 2009. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*.

Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H. Chi, Nathanael Schärli, and Denny Zhou. 2023. Large language models can be easily distracted by irrelevant context. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 31210–31227. PMLR.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 36, pages 8634–8652.

Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. 2021. ALFWorld: Aligning text and embodied environments for interactive learning. In *International Conference on Learning Representations (ICLR)*.

Mudit Verma, Siddhant Bhambri, and Subbarao Kambhampati. 2024. On the brittle foundations of ReAct prompting for agentic large language models. *Preprint*, arXiv:2405.13966.

Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2024. Voyager: An open-ended embodied agent with large language models. *Transactions on Machine Learning Research*.

Zora Zhiruo Wang, Jiayuan Mao, Daniel Fried, and Graham Neubig. 2025. Agent workflow memory. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 63897–63911. PMLR.

Albert Webson and Ellie Pavlick. 2022. Do prompt-based models really understand the meaning of their prompts? In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 2300–2344, Seattle, United States. Association for Computational Linguistics.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, et al. 2025. Qwen3 technical report. *Preprint*, arXiv:2505.09388.

Yifan Yang, Ziyang Gong, Weiquan Huang, Qihao Yang, Ziwei Zhou, Zisu Huang, Yan Li, Xuemei Gao, Qi Dai, Bei Liu, Kai Qiu, Yuqing Yang, Dongdong Chen, Xue Yang, and Chong Luo. 2026. SkillOpt: Executive strategy for self-evolving agent skills. *Preprint*, arXiv:2605.23904.

Shunyu Yao, Howard Chen, John Yang, and Karthik Narasimhan. 2022. WebShop: Towards scalable real-world web interaction with grounded language agents. In *Advances in Neural Information Processing Systems*, volume 35, pages 20744–20757.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*.

Qizheng Zhang, Changran Hu, Shubhangi Upasani, Boyuan Ma, Fenglu Hong, Vamsidhar Kamanuru, Jay Rainton, Chen Wu, et al. 2026. Agentic context engineering: Evolving contexts for self-improving language models. In *International Conference on Learning Representations (ICLR)*.

Andrew Zhao, Daniel Huang, Quentin Xu, Matthieu Lin, Yong-Jin Liu, and Gao Huang. 2024. ExpeL: LLM agents are experiential learners. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(17):19632–19642.

Zihao Zhao, Eric Wallace, Shi Feng, Dan Klein, and Sameer Singh. 2021. Calibrate before use: Improving few-shot performance of language models. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 12697–12706. PMLR.

A Implementation Details

Models. The GPT-4o-mini alias resolved to the snapshot of 2024-07-18, and GPT-4.1-mini was called as the snapshot of 2025-04-14. Local models were served with vLLM in bf16 with a context of 16,384 tokens, with `enable_thinking=false` for Qwen3. Evaluation calls used at most 256 completion tokens, and distillation calls at most 512.

Estimators. For the between-pool standard deviation, we average $((R_i - R_j)^2 - v_{ij})/2$ over replicate pairs within each source-distiller-reader cell and then across cells, truncate the pooled variance at zero, and report its square root; v_{ij} is the task-bootstrap variance with a small-sample correction.

Pools. Demonstration pairs are selected by seeds 0, 1 and 5. Trajectories are balanced at six per task type except one for heat, because GPT-4o-mini solved only one heat game, 31 in all. Pools are distilled with one call per task type at temperature

0, from a prompt that asks for 3 to 6 concise general rules, each one imperative line with no object numbers or room layouts. Generic rules use the same instruction with a fixed description of the environment in place of trajectories, and each is then rewritten to the matched item's word count. The focal control needed the most placeholder tokens, 30 of 472 body words; the crossover controls needed none.

Artifacts. The repository on the first page holds the harness (`skillrc/`), configurations (`configs/`), pools and result summaries (`data/`) and analysis scripts (`scripts/analysis/`). The crossover is under `configs/w1_2026_10_13` and `data/summaries/w1a_2026-10-13`. The same-window configurations and summaries are not yet public; that study uses replicate 1 of each source, the Qwen3-32B distillations for the three-pool comparisons and both distillations for Llama-3.1-8B's six pools, with the same demonstration seeds. Figure icons are from Twemoji (CC BY 4.0).